VENDOR PERFORMANCE
ANALYSIS REPORT

A data-driven review of sales, purchasing, profitability and inventory

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Project	Vendor Performance Analysis

Executive Summary

This project analyses vendor and brand performance using purchasing, sales, profitability and inventory-related measures. The work combines data preparation and analysis in Python/SQL with an interactive Power BI dashboard. The dashboard is designed to help a business compare vendor contribution, identify leading brands, review stock turnover and monitor unsold inventory.

Dashboard indicator	Reported value	Interpretation
Total purchase	307.34M	Overall purchasing value represented in the dashboard.
Total sales	441.41M	Overall sales value represented in the dashboard.
Gross profit	134.07M	Difference between sales and purchase values as represented by the dashboard measure.
Profit margin	38.72	Displayed as the average of ProfitMargin; unit/format should be verified before treating it as a percentage.
Unsold capital	2.71M	Value represented by the UnsoldInventoryValue measure.
Top 10 vendor purchase contribution	201.89M (~65.7%)	Top ten vendors' purchase value as a share of total purchase.

Overall takeaway Sales exceed purchases in the dashboard totals, while the top ten vendors account for approximately 65.7% of total purchasing value. This indicates a concentrated purchasing base that merits ongoing monitoring.

The report documents the dashboard's observed results and explains how the visuals can support vendor review. It also records important interpretation limits, particularly where the dashboard uses aggregated fields or numeric brand identifiers.

01 | PROJECT CONTEXT

Introduction and Project Objectives

Understanding why vendor-level measurement matters to purchasing and inventory decisions.

1.1 Introduction

Vendor performance analysis is the structured assessment of suppliers using operational and financial indicators. A business may purchase from many vendors and sell products across multiple brands. Without a consolidated view, it can be difficult to see which vendors contribute most to purchasing, which brands generate sales, and where inventory may be moving slowly.

This project develops a data-analysis workflow and a Power BI dashboard to bring these measures together. The dashboard presents headline indicators as KPI cards and provides supporting visuals for vendor contribution, sales rankings, brand sales and stock turnover.

1.2 Business problem

The central problem addressed is the need to convert transaction and inventory-related data into a concise performance view. Stakeholders need to compare vendors and brands, monitor profitability, and identify areas that may require further investigation. Raw tables alone make these comparisons less accessible, particularly when multiple measures must be considered together.

1.3 Project objectives

- Consolidate relevant vendor, brand, purchase, sales and inventory fields into an analysis-ready dataset.
- Calculate or use business measures such as gross profit, profit margin, stock turnover and unsold inventory value.
- Summarise total purchasing, sales, gross profit, profit margin and unsold capital through KPI cards.
- Identify the leading vendors by sales and the leading brands by sales.
- Compare vendor purchasing contribution and examine stock-turnover patterns.
- Present the results in an interactive Power BI dashboard that is easy to review.

1.4 Scope

The report focuses on the measures and visuals currently present in the Power BI dashboard. It does not claim to evaluate supplier delivery reliability, contractual compliance, product quality or service levels because those outcomes are not represented in the dashboard figures shared for this report. The analysis is descriptive: it highlights patterns for review rather than proving the causes of vendor or brand performance.

How to use the dashboard Start with the KPI cards for the overall position, then use the vendor and brand visuals to investigate which entities contribute to the totals and which may warrant closer review.

02 | DATA AND APPROACH

Dataset, Tools and Methodology

A reproducible workflow from source data to an interactive reporting layer.

2.1 Dataset overview

The project dataset contains vendor- and brand-related records with financial and operational fields. The fields visible in the Power BI model include Brand, VendorName, ActualPrice, FreightCost, GrossProfit, OrderSize, ProfitMargin, PurchasePrice, SalesToPurchaseRatio, StockTurnover, TotalExciseTax, TotalPurchase-related measures, TotalSalesDollars, TotalSalesPrice and TotalSalesQuantity. The analysis-ready table used in the project is referred to as vendor_sales_summary in the preparation workflow.

2.2 Tools and technologies

Tool	Role in the project
Python	Data ingestion, preparation, feature creation and exploratory analysis.
Pandas and NumPy	Tabular data handling and numerical operations.
SQLite / SQL	Storing and querying the prepared vendor summary data.
Jupyter Notebook	Running and documenting the analysis workflow.
Power BI Desktop	Building KPI cards, charts, filters and the final dashboard.

2.3 Data preparation and measures

The preparation workflow creates a consolidated vendor summary from the source tables and derives additional measures used for analysis. The notebook workflow includes a filtered dataset that retains records with positive GrossProfit, positive ProfitMargin and positive TotalSalesQuantity. This filter is important when interpreting results: records excluded by these conditions are not represented in the filtered analysis dataset.

Measure / field	Analytical purpose
GrossProfit	Represents the gross profit measure used in the dashboard.
ProfitMargin	Supports comparison of profitability relative to sales or another defined base; formula and unit should be documented.
StockTurnover	Used to examine how quickly inventory is moving.
UnsoldInventoryValue	Represents capital tied up in inventory that has not been sold.
SalesToPurchaseRatio	Provides a comparison between sales and purchasing activity.

2.4 Analysis and visualisation workflow

- Load and consolidate the relevant source data into a vendor-level summary.
- Apply the project's data filters and create derived measures.
- Explore vendor and brand performance using grouped summaries and rankings.
- Export the final analysis dataset to CSV for Power BI.
- Import the CSV and build KPI cards, a purchase-contribution donut, ranking charts and a stock-turnover matrix.
- Apply visual-level filters such as Top 10 or Bottom 10 where appropriate, then arrange the dashboard for review.

Methodology note The report uses the dashboard values supplied during the project discussion. The exact source-table row count, date range, currency, and formal calculation definitions were not available, so they are intentionally not invented here.

03 | RESULTS

KPI Analysis and Key Findings

The dashboard's headline indicators provide a compact view of financial and inventory performance.

3.1 Overall financial indicators

KPI	Dashboard value	Analysis
Total Purchase	307.34M	Represents the total purchasing value in the dashboard.
Total Sales	441.41M	Sales are higher than purchases by 134.07M in the displayed totals.
Gross Profit	134.07M	The displayed gross profit equals the arithmetic difference between total sales and total purchase.
Profit Margin	38.72	The card uses Average of ProfitMargin. Confirm whether the source values are stored as ratios or percentages and whether an average is the intended aggregate.
Unsold Capital	2.71M	Shows the value associated with unsold inventory; this can help prioritise stock review.

3.2 Derived comparison

Using the displayed totals, the difference between total sales and total purchase is 441.41M − 307.34M = 134.07M, matching the Gross Profit card. This arithmetic consistency is a useful dashboard check. It does not by itself confirm that all costs, discounts, returns, taxes or other accounting adjustments are included in the gross-profit definition.

The top ten vendors' purchase contribution is displayed as 201.89M. Compared with total purchase of 307.34M, this is approximately 65.7%. The calculation is 201.89 ÷ 307.34 × 100. Therefore, around two-thirds of the purchasing value represented in the dashboard is concentrated among the ten leading vendors by purchase value.

3.3 Interpretation of unsold capital

The unsold capital KPI is 2.71M. This indicates that a measurable amount of value is represented by inventory that has not yet been sold. The figure should be interpreted alongside stock age, demand, product category and replenishment lead time before any purchasing or clearance decision is made. The current dashboard provides the value measure but does not show ageing bands or inventory holding duration.

3.4 KPI quality checks

- Confirm the currency and scale represented by the M suffix and show the currency consistently in the report and dashboard.
- Verify the ProfitMargin calculation and format; the card currently displays 38.72 as Average of ProfitMargin without a visible percent sign.

- Confirm whether GrossProfit is defined as TotalSalesDollars minus TotalPurchaseDollars or whether other costs are included.
- Confirm that UnsoldInventoryValue is aggregated using Sum and that duplicate inventory records are not counted more than once.

Key finding The dashboard indicates 441.41M in sales, 307.34M in purchases and 134.07M in gross profit, alongside 2.71M in unsold capital. These are the headline figures for management review.

04 | ENTITY PERFORMANCE

Vendor and Brand Performance Analysis

Using rankings and operational indicators to focus attention on the entities that matter most.

4.1 Vendor sales performance

The Top 10 Vendors by Sales bar chart ranks vendors by TotalSalesDollars. The leading vendors visible in the dashboard include Diageo North America Inc, Martignetti Companies, Pernod Ricard USA, Jim Beam Brands Company, Bacardi USA Inc, Constellation Brands Inc, E & J Gallo Winery, Brown-Forman Corp, Ultra Beverage Company LLP and M S Walker Inc. These vendors form the leading sales group shown in the current dashboard.

This ranking helps identify vendors associated with the highest sales value in the prepared dataset. A high sales ranking should not automatically be interpreted as the best overall supplier: purchase cost, gross margin, stock availability, delivery performance and product quality would also be relevant to a broader supplier scorecard.

4.2 Purchase contribution

The purchase-contribution donut chart focuses on the ten vendors with the largest purchase values. The centre of the chart displays 201.89M, equivalent to approximately 65.7% of total purchase of 307.34M. The chart therefore communicates purchasing concentration and can help a reviewer understand the organisation's reliance on its major suppliers.

4.3 Brand sales performance

The Top 10 Brands by Sales chart ranks brands using TotalSalesDollars. In the current Power BI model, the Brand field appears as numeric identifiers rather than descriptive brand names. The ranking is still useful for comparing sales values, but the business interpretation is limited until those identifiers are mapped to readable brand names.

The Bottom 10 Brands by Sales chart highlights the ten brands with the lowest sales values among the records included in the visual's filter context. This can be used as a screening list for further review. Low sales alone do not prove poor performance: a brand may be new, seasonal, premium, or intentionally stocked in small quantities.

4.4 Vendor stock turnover

The Vendor Stock Turnover Analysis matrix presents VendorName alongside StockTurnover. This provides a vendor-level comparison of the turnover measure. The matrix should be checked to ensure that StockTurnover is aggregated using the intended method. In the current screenshot,

the value header appears as Sum of StockTurnover; for a ratio-like measure, an average or a properly recalculated ratio may be more meaningful than a sum, depending on how the source measure is defined.

Interpretation caution The report does not label a vendor or brand as definitively poor based only on a ranking. The charts identify candidates for investigation; purchasing context, demand, margin and stock age should be considered before action.

05 | ACTION AND CONCLUSION

Recommendations, Limitations and Conclusion

Turning dashboard observations into practical next steps.

5.1 Recommendations

Review supplier concentration: Because the top ten vendors represent approximately 65.7% of purchase value, periodically review supplier dependency, alternative sourcing options and continuity risks.

Investigate low-turnover inventory: Use stock-turnover results to identify products or vendor portfolios that may be moving slowly. Validate with stock age, demand forecasts and seasonality before changing order quantities.

Review unsold capital: Break the 2.71M unsold inventory value into age bands and product categories. Prioritise items with long holding periods or weak recent demand.

Improve brand identification: Create a mapping from numeric Brand IDs to readable brand names so that the Top 10 and Bottom 10 brand charts can be interpreted by business users.

Standardise KPI definitions: Document the formulas, units and aggregation rules for GrossProfit, ProfitMargin, StockTurnover and UnsoldInventoryValue.

Extend the dashboard: If data is available, add vendor delivery reliability, purchase order fulfilment, return rates, product quality and time-based trends for a more complete supplier scorecard.

5.2 Limitations

- The report is based on the dashboard figures and visible field names shared during the project. The full raw dataset and data dictionary were not supplied for independent validation.
- The reporting period and currency were not specified; values are therefore reproduced using the dashboard's M notation without adding a currency symbol.
- ProfitMargin is shown as Average of ProfitMargin with a value of 38.72. Its exact formula, storage scale and correct display format require confirmation.
- Brand values appear as numeric identifiers, which limits descriptive interpretation.
- The vendor stock-turnover matrix appears to use Sum of StockTurnover; the appropriate aggregation should be verified against the measure definition.
- The analysis is descriptive and does not establish causal reasons for high or low sales, profit or stock turnover.

5.3 Conclusion

The Vendor Performance Analysis project demonstrates how Python, SQL, Jupyter Notebook and Power BI can be combined to transform operational data into a practical reporting dashboard. The dashboard brings together purchasing, sales, gross profit, profit margin and unsold inventory indicators, while allowing users to compare vendor and brand performance.

The key dashboard observations are that total sales are 441.41M, total purchase is 307.34M, gross profit is 134.07M and unsold capital is 2.71M. The top ten vendors account for approximately 65.7% of total purchasing value. These findings provide a starting point for supplier concentration reviews, inventory investigation and more detailed profitability analysis.

Final conclusion The dashboard is a useful decision-support tool and a strong foundation for a broader vendor scorecard. Its value will improve further when KPI definitions, brand labels, aggregation rules and reporting periods are documented and validated.

Appendix: Dashboard evidence

